Week #12: 16 Nov - 20 Nov

Subject/:الموضوع Chemistry

Unit Title: عنوان الوحدة Unit 4: Oxidation and reduction

Grade الصف: 10

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: Batteries	I can EXPLAIN how primary and secondary batteries use redox reactions to convert chemical energy into electrical energy and EVALUATE their suitability for different uses.	**L1 — Batteries:** diagram/MCQ check and compare-and-justify task. **L1 differentiation:** MILD—identify; HOT—explain/compare; EXTRA HOT—evaluate a UAE battery choice. **L2 — Formative:** Unit 4 checkpoint (MCQ, application and reasoning) with feedback. **L2 — GRASP final check:** rubric review of completion, scientific evidence, accuracy and presentation. **L2 differentiation:** MILD—guided checklist; HOT—self/peer review; EXTRA HOT—justify final edits. **UDL:** visuals, vocabulary prompts, exemplars and accessible feedback.	**Lesson 1:** McGraw-Hill Unit 4 Batteries Science Notebook, lesson PPT and battery comparison visuals. **Lesson 2:** Unit 4 revision/formative resources; GRASP brief, checklist and task-specific rubric. **Access:** Microsoft Teams class materials and Toddle class drive (no new external link required).
Lesson 2: Formative Assessment & GRASP Final Check	I can APPLY my Unit 4 knowledge in a formative assessment and EVALUATE my GRASP project against the task requirements and rubric before final submission.		
Homework الواجبات	Review batteries, redox reactions and voltaic cells using McGraw-Hill and Teams materials. Complete any outstanding GRASP sections, apply feedback and prepare the project for final submission as instructed.		
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Week 12 reminder: Students will complete a Unit 4 formative assessment and a final GRASP progress/quality check. Please ensure they bring their current project work and review the rubric and teacher feedback. Final submission arrangements will be confirmed through Toddle/Teams. Hello Parents, Please encourage your child to review and practice Science regularly each week using the resources available in their class Microsoft Teams - class material folder which is organized by week. The presentations, readings, videos, practice activities, etc. posted in Teams are the primary learning materials and should be used throughout the unit to support understanding and achievement. Science study guides will provide basic assessment information, topics, and expectations; however, they are intended as a guide only and will not contain detailed study materials, these are found in the class teams. Effective preparation for graded formatives and summative requires students to actively engage with the learning materials provided in their Toddle class drive. Thank you for partnering with us to support your child's learning and success. Ms. Stacey (Science Coordinator)		

Technology Integration دمج التكنولوجيا	Battery diagrams and short digital models illustrate redox reactions, electron flow and rechargeability. Teams/Toddle provide the formative revision materials, GRASP rubric, project evidence and feedback.
Cross-curricular links الروابط بين المواد الدراسية	**Physics:** current, potential difference and energy transfer. **Mathematics:** interpreting battery data and making evidence-based comparisons. **Environmental Science:** battery waste, recycling and sustainable energy use in the UAE.
External Assessment Integration دمج التقييمات الخارجية	MAP / CAT4 skills: interpreting scientific data, using evidence, applying concepts and explaining conclusions; classroom formative checks inform next steps.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	**Thinking:** compare battery technologies and justify scientific decisions. **Self-management:** organize GRASP evidence, meet checkpoints and prioritize revisions. **Communication:** explain reasoning and use rubric language in feedback. **Collaboration & wellbeing:** respectful peer feedback, reflection and manageable next steps.
Standards المعايير	HS-PS1-2: Construct and revise explanations of chemical reactions using electron states and periodic trends. Unit 4 application: explain electron transfer in batteries; compare rechargeable and non-rechargeable systems; evaluate evidence and communicate justified scientific conclusions.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How can evidence from redox chemistry help us choose suitable batteries, and how can formative feedback improve the quality of a scientific GRASP project?
Success Criteria معايير النجاح **(Measurable)**	**L1 — BATTERIES** **MUST (DOK 1–2):** IDENTIFY anode, cathode, electron flow and primary/secondary batteries. **SHOULD (DOK 3):** EXPLAIN battery operation through redox reactions and COMPARE battery uses. **COULD (DOK 4):** EVALUATE and JUSTIFY a battery choice for a UAE application using evidence. **L2 — FORMATIVE & GRASP FINAL CHECK** **MUST (DOK 1–2):** RECALL Unit 4 concepts and IDENTIFY completed/missing GRASP requirements. **SHOULD (DOK 3):** APPLY redox knowledge and ANALYZE project quality against the rubric. **COULD (DOK 4):** JUSTIFY revisions and PRIORITIZE final improvements based on feedback.
National Identity Domain مجالات الربط بالهوية الوطنية	Conservation | Global Understanding | UAE Values & Sustainable Development Connect responsible battery selection, safe disposal and recycling to UAE sustainability priorities. Use GRASP reflection to consider how scientific decisions benefit UAE communities and protect the environment.